

# AI-Based Industrial Maintenance and Process Optimization System

## MILESTONE 3 REPORT

*Advanced User Interface and Operator Experience*

Domain: Artificial Intelligence

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Team: Team D

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## TEAM MEMBERS

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## Document purpose

This is a standalone milestone 3 report. It documents the advanced frontend interface, operator experience enhancements, and comprehensive user interaction features delivered for this milestone. Building on the API-driven backend and data management infrastructure from Milestone 2, this milestone focuses on delivering an intuitive, feature-rich web-based operations console. It intentionally contains no screenshots because the working application will be demonstrated separately.

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## 1. MILESTONE 3 AT A GLANCE

| Area            | Milestone 3 delivery                                                          |
|-----------------|-------------------------------------------------------------------------------|
| Milestone theme | Multi-page web interface with interactive dashboards and operator experience  |
| Primary users   | Operations staff, maintenance engineers, plant supervisors viewing dashboards |
| Main inputs     | API responses from backend (machines, telemetry, predictions, incidents)      |
| Main outputs    | Interactive web dashboards, visualizations, CSV exports, status updates       |

Milestone 3 transforms the FactoryOps AI platform into an interactive, operator-friendly web console. Building on the robust backend API and data management capabilities from Milestone 2, this milestone delivers a sophisticated Streamlit-based user interface that converts raw operational data into actionable insights. The frontend provides operators with real-time visibility into fleet condition, machine health, failure predictions, incident management, and detailed telemetry analysis through an intuitive multi-page dashboard. The emphasis is on user experience, data accessibility, and decision support without requiring direct database interaction.

## 2. OBJECTIVES AND SCOPE

### 2.1 Objectives

- Build a responsive, modern web interface for the FactoryOps AI monitoring platform.
- Provide operators with real-time visibility into machine health, fleet status, and operational metrics.
- Deliver advanced data visualization and charting capabilities for telemetry inspection and trend analysis.
- Implement role-based access control supporting multiple user personas (Admin, Operator, Maintenance).
- Create a multi-page application with specialized views for different operational domains.
- Enable data export capabilities for integration with external systems and reporting.
- Design an intuitive command center that prioritizes critical information and actionable alerts.

2.2 Scope boundary

| Scope decision           | Details                                                                                                                                    |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Included in Frontend     | Streamlit multi-page dashboard, Plotly visualizations, CSV exports, responsive UI design, role-based access, dark theme operations console |
| Not claimed as completed | Mobile-native app, real-time WebSocket streaming, advanced drill-down charts, production authentication                                    |
| Reason for boundary      | Frontend focuses on intuitive operator experience using Streamlit. Backend API and ML components are from Milestone 2.                     |

3. DELIVERED SYSTEM ARCHITECTURE

The frontend is organized as a modular Streamlit application with clear separation of concerns:

User Login → Dashboard Navigation → Page Selection → API Data Fetch → Data Transformation  
→ Interactive Visualization → Operator Action

| Frontend Component      | Purpose & Implementation                                                                       |
|-------------------------|------------------------------------------------------------------------------------------------|
| Authentication Page     | Demo login form with role selection (Admin, Operator, Maintenance); session state management   |
| Command Center          | Real-time dashboard with KPI metrics, machine status cards, risk visualization, decision queue |
| Fleet Explorer          | Machine inventory with charts (pie, bar), register table, inline status update form            |
| Predictions & Incidents | Scatter plot for failure probability, immediate attention list, kanban incident board          |
| Telemetry Lab           | Machine selector, metric cards, multi-signal time-series chart, raw data table                 |

## 4. IMPLEMENTED FRONTEND MODULES

| Module                  | File                     | Key UI Elements                                                                                  |
|-------------------------|--------------------------|--------------------------------------------------------------------------------------------------|
| Authentication          | app.py                   | Login form, role selector, session state management, navigation sidebar                          |
| Command Center          | command_center.py        | Metric cards (5 KPIs), machine condition grid, risk radar bar chart, decision queue list         |
| Fleet Explorer          | fleet_explorer.py        | Status pie chart, location bar chart, machine register table, status update form, CSV download   |
| Predictions & Incidents | predictions_incidents.py | Failure probability scatter plot, top 8 predictions list, incident kanban board, status cards    |
| Telemetry Lab           | telemetry_lab.py         | Machine selector dropdown, metric cards (5 sensors), time-series line chart, signal multi-select |
| Utilities               | utils.py                 | Metric card renderer, status badges, chart layouts, table formatter, state helpers               |

## 5. DASHBOARD PAGES AND FEATURES

| Dashboard Page          | Key Metrics & Visualizations                                                                                      | User Actions Available                                             |
|-------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Command Center          | Fleet availability, avg health, critical count, watch list, open incidents, machine cards (12), risk radar top 10 | View dashboards, filter by condition, monitor alerts               |
| Fleet Explorer          | Status distribution pie, location bar chart, machine register table (10 columns)                                  | Filter machines, update status, export CSV, sort table             |
| Predictions & Incidents | Failure probability scatter plot, immediate attention table (8 rows), incident kanban (4 columns)                 | View predictions, sort by risk, track incidents, export CSV        |
| Telemetry Lab           | Machine selector, 5 metric cards, time-series line chart (multi-signal), raw readings                             | Select machine, pick signals, inspect trends, export telemetry CSV |

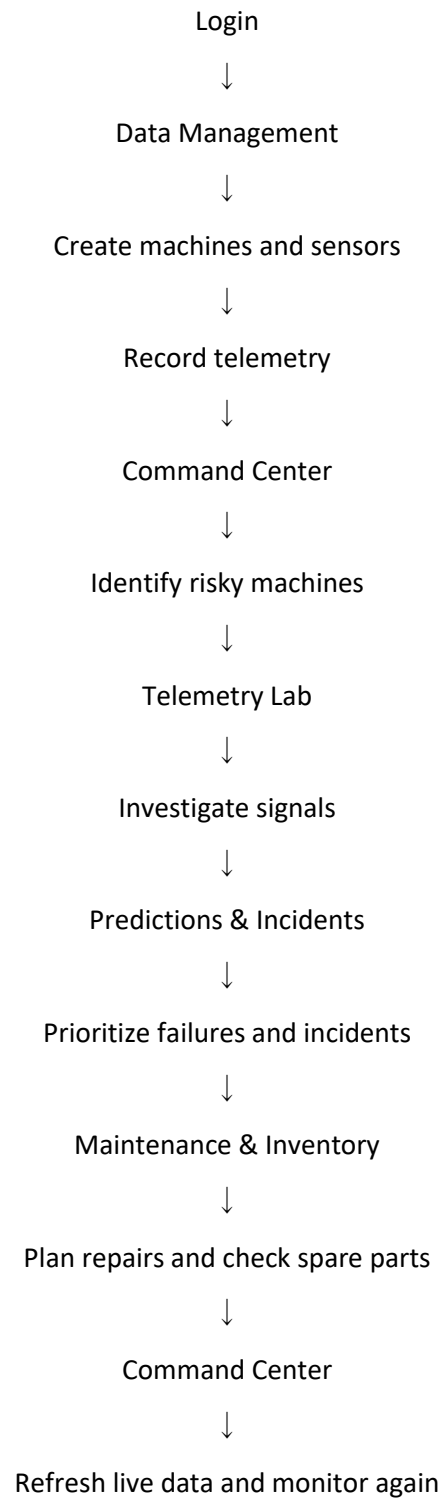
|                        |                                                                                             |                                                               |
|------------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------|
|                        | table                                                                                       |                                                               |
| <b>Data Management</b> | CRUD forms for machines, sensors, telemetry, predictions, maintenance, incidents, inventory | Create records, edit details, delete entries, bulk operations |

## 6. USER INTERFACE COMPONENTS

### 6.1 Visual Design System

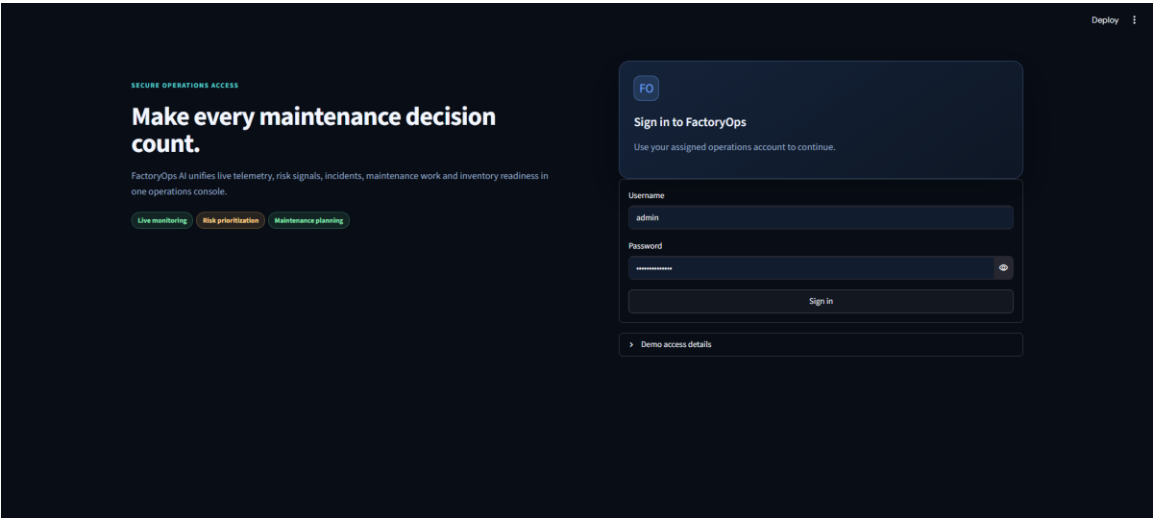
| UI Component              | Usage & Styling                                                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Metric Cards</b>       | Display KPIs with label, value, detail text; used in Command Center (5x) and Telemetry Lab (5x); light border on dark background                   |
| <b>Machine Cards</b>      | Grid layout (3 columns) showing machine name, location, temp, failure probability; left border color-coded by condition (green/amber/red)          |
| <b>Status Badges</b>      | Pill-shaped badges for condition (Normal/Warning/Critical) and status (Running/Idle/Maintenance/Stopped); color-coded                              |
| <b>Charts (Plotly)</b>    | Pie charts (status distribution), bar charts (location count, risk radar), scatter plots (health vs. failure prob), line charts (telemetry trends) |
| <b>Tables (Streamlit)</b> | Interactive dataframes with sortable columns, formatted numbers (percentages, decimals), color highlighting, CSV export buttons                    |
| <b>Forms</b>              | Status update form with selectbox and submit button; login form with username/password fields; data entry forms for CRUD operations                |

## 6.2 USER FLOW



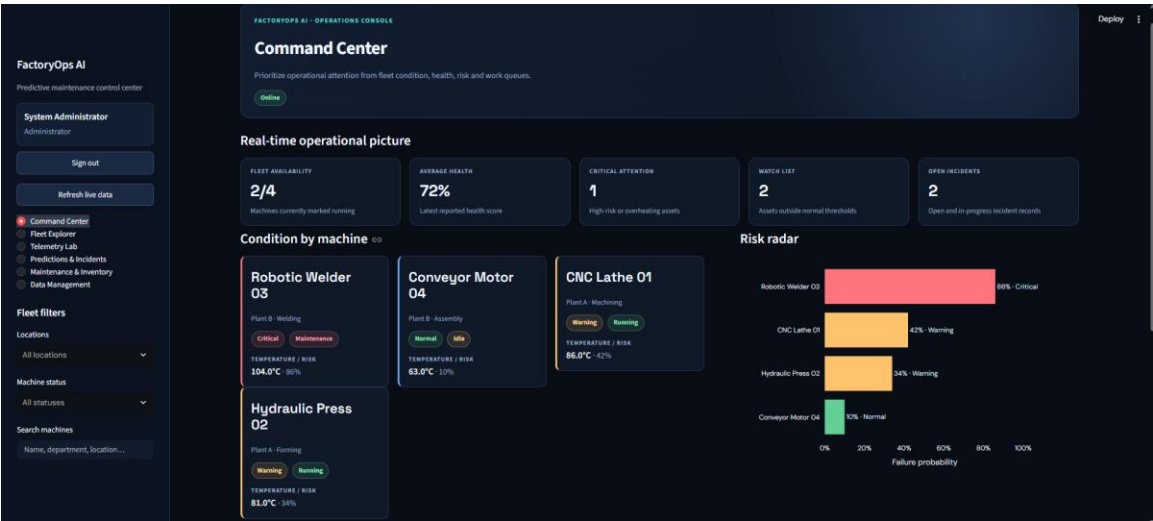
## Login Page

Securely access the FactoryOps AI operations dashboard.  
Users sign in to monitor machines, telemetry, predictions, and maintenance activities.



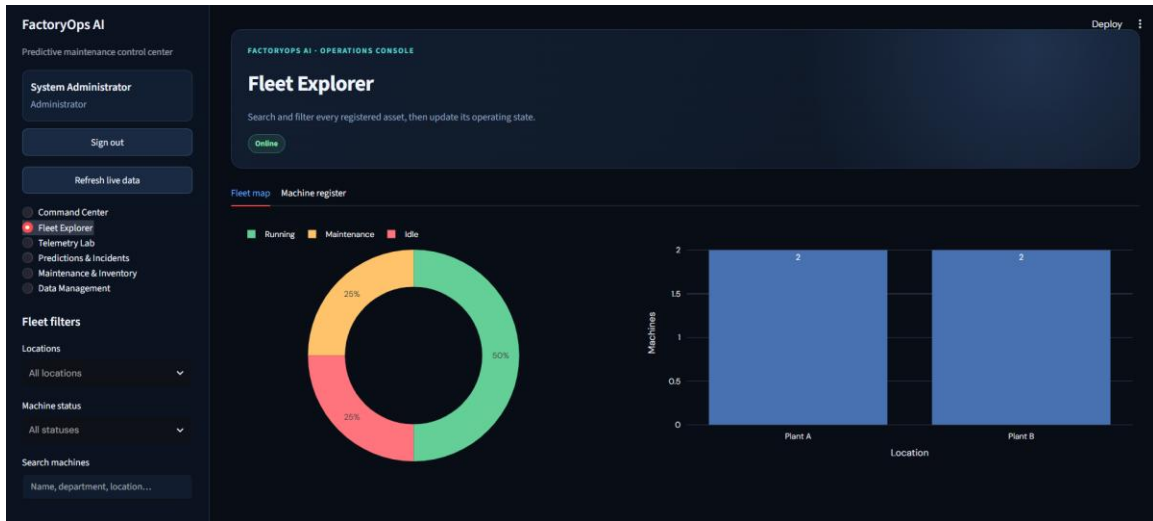
## Command Center

Provides a real-time overview of overall factory and machine health.  
Highlights risks, critical assets, open incidents, and maintenance priorities.



## Fleet Explorer

Displays all registered machines with their status, location, health, and risk levels. Users can filter, search, export, and update machine operating status.



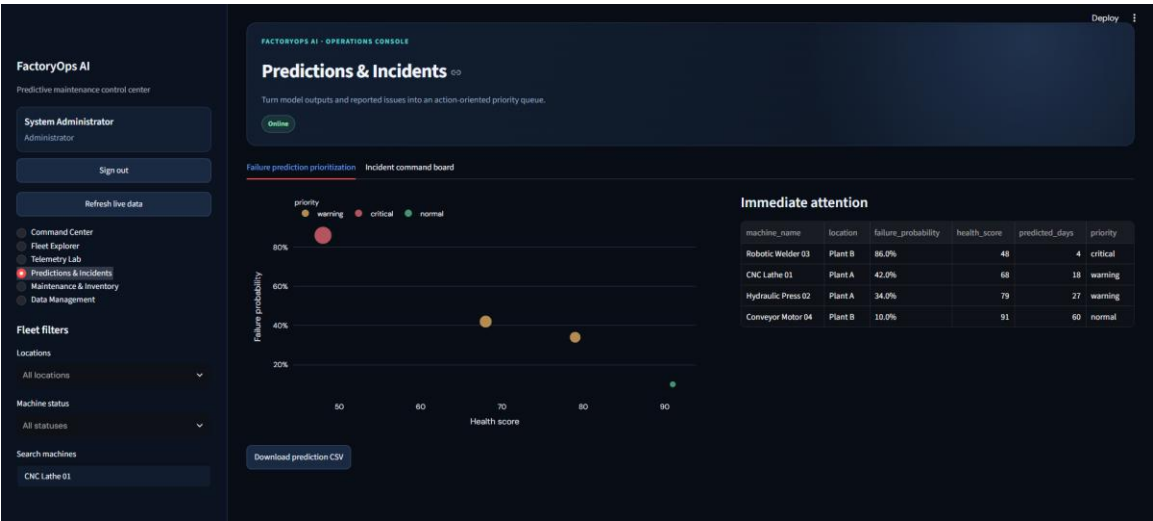
## Telemetry Lab

Visualizes machine sensor readings such as temperature, vibration, power, and RPM. Helps users investigate abnormal machine behavior through charts and raw data.



## Prediction & Incidents

Shows machine failure probabilities, health scores, and predicted failure timelines.  
Organizes incidents by priority and status so users can take timely action.



## Data Management

Allows users to create and manage machines, sensors, telemetry, incidents, and maintenance records. The entered data is stored in the backend database and powers the dashboard insights.

The screenshot shows the 'Data Management' dashboard in the FactoryOps AI interface. The left sidebar is identical to the previous screenshot, with the 'Data Management' link now active.

The main content area is titled 'Data Management' and includes a sub-header 'Create the operational records that power the dashboard through the FastAPI service.' Below this, there is a 'Submit records directly through the FastAPI API. Required fields match the current backend schemas.' note.

Below the note are tabs for 'Machine', 'Telemetry', 'Maintenance', 'Incident', 'Inventory', 'Notification', and 'Sensor'. The 'Machine' tab is active.

The 'Machine' tab contains a form with the following fields:

- 'Machine name \*' (text input)
- 'Department' (text input)
- 'Location' (text input)
- 'Status' (dropdown menu with 'Running' selected)

Below the form is an 'Add machine' button.

## 7. DATA VISUALIZATION AND CHARTS

| Chart Type                         | Where Used                                                                     | Data Encoding                                                                                       |
|------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Pie Chart (Donut)</b>           | Fleet Explorer - Status distribution                                           | Slices = status types; colors = green/amber/red; center text = machine count                        |
| <b>Bar Chart</b>                   | Fleet Explorer (location), Command Center (risk radar), Predictions (severity) | X-axis = category, Y-axis = count/probability; colors encode status; labels show values             |
| <b>Scatter Plot</b>                | Predictions page - Health vs. Failure Probability                              | X = health score, Y = failure probability, size = bubble magnitude, color = risk level              |
| <b>Line Chart</b>                  | Telemetry Lab - Multi-signal time-series                                       | X = record ID, Y = sensor value, lines = different signals, interactive legend for signal filtering |
| <b>Horizontal Bar (Risk Radar)</b> | Command Center - Top 10 at-risk machines                                       | Bar length = failure probability %, color-coded by condition, labels show machine names and risk %  |

## 8. USER AUTHENTICATION AND ACCESS CONTROL

| User Role            | Dashboard Access                                                           | Actions Allowed                                                                   |
|----------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Administrator</b> | All pages (Command Center, Fleet, Predictions, Telemetry, Data Management) | View, create, edit, delete all data; manage users; export CSV                     |
| <b>Operator</b>      | Command Center, Fleet Explorer, Predictions, Telemetry Lab only            | View dashboards, export CSV data; NO create/edit/delete permissions               |
| <b>Maintenance</b>   | Fleet Explorer, Predictions, Telemetry Lab, Data Management (partial)      | View data, update maintenance records, manage incidents; limited telemetry access |

## 9. TECHNOLOGY STACK

| Frontend Technology            | Purpose in Dashboard                                                                      |
|--------------------------------|-------------------------------------------------------------------------------------------|
| Streamlit 1.x                  | Core framework for multi-page web app; handles routing, session state, layout components  |
| Plotly Express & Graph Objects | Interactive charting library; pie, bar, scatter, line charts with hover, zoom, export     |
| Pandas DataFrames              | Data transformation, filtering, sorting; powers tables and chart data preparation         |
| Python requests                | HTTP client to fetch JSON from FastAPI backend; error handling and timeouts               |
| CSS3 (custom theme)            | Dark-mode operations console styling; color variables, responsive grid, component styling |
| Google Fonts                   | Typography: DM Sans (body), Space Grotesk (headers) for professional operations look      |
| Streamlit session_state        | Maintains user login, role, navigation state across page interactions                     |

## 10. VERIFICATION APPROACH

Verification for this milestone focuses on user interaction, data flow, and visual presentation:

| Frontend Check            | Expected Evidence                                                                                                               |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Login page loads          | Role selection dropdown and username/password fields render correctly                                                           |
| Authentication works      | Valid credentials establish session; invalid credentials show error; session persists across pages                              |
| Role-based access         | Operator cannot access Data Management; Maintenance sees limited telemetry; Admin sees all pages                                |
| Command Center displays   | All 5 metric cards populate; machine grid shows 12 cards sorted by condition; risk radar shows top 10 machines                  |
| Fleet Explorer works      | Pie chart shows status distribution; bar chart shows locations; machine table loads all columns; status update form works       |
| Predictions page displays | Scatter plot renders with health/probability axes; immediate attention list shows top 8; incident kanban shows 4 status columns |
| Telemetry Lab functions   | Machine selector dropdown works; metric cards show latest values; time-series chart renders; signal multi-select filters chart  |
| CSV exports               | Download buttons generate valid CSV files with headers and data; files open correctly in Excel                                  |
| Charts are interactive    | Hover tooltips show data; zoom/pan work on line charts; color encoding matches status rules                                     |
| Responsive layout         | Dashboard adapts to different screen widths; no horizontal scrolling; columns reflow appropriately                              |
| Error handling            | Missing/empty data shows "no data" messages; API errors display gracefully; forms validate input                                |
| Dark theme renders        | Background is #080d16; text is readable; accent colors (green/amber/red) display correctly                                      |

## 10.1 Demonstration commands

15. Backend: `uvicorn backend.api.main:app --reload`

16. Dashboard: `streamlit run app.py`

## 11. CURRENT LIMITATIONS AND RISK CONTROLS

| Frontend Limitation           | Risk Control / Mitigation                                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Demo authentication           | Hardcoded credentials in DEMO_USERS dict. Control: Use environment variables to override; deploy OAuth 2.0 before production |
| No real-time updates          | Dashboard updates only when user clicks or page reloads. Control: Add WebSocket or polling in Milestone 4                    |
| In-memory session state       | Reloading page loses login state. Control: Implement Redis or database-backed sessions for multi-user production             |
| Single-machine telemetry view | Telemetry Lab shows one machine at a time. Control: Add multi-machine comparison in next milestone                           |
| Limited chart interactivity   | Charts show hover tooltips but no drill-down. Control: Plan Plotly event handlers for advanced interactions                  |
| No data filtering             | Tables show all records; no search or filter. Control: Add Streamlit column filters and search fields in Milestone 4         |

## 12. NEXT MILESTONE PLAN

Milestone 4 will enhance the frontend with advanced interactivity, real-time updates, and expanded analytics:

- Add real-time WebSocket updates for live dashboard refresh
- Implement advanced filtering and search across all dashboard tables
- Build executive dashboard with KPI trends, historical analytics, and forecasting
- Develop mobile-responsive design for on-floor tablet access
- Add drill-down and cross-filtering between dashboard charts
- Integrate user preferences (saved filters, custom themes, alerts)
- Implement audit logging showing who viewed/modified what data and when

## 13. CONCLUSION

Milestone 3 delivers a sophisticated, operator-friendly web interface that transforms the FactoryOps AI platform from an API-centric backend into an accessible, interactive operations console. The Streamlit-based frontend provides operators with comprehensive visibility into fleet health, machine status, predictive analytics, and incident management through intuitive multi-page navigation and advanced data visualization.

By combining real-time data access from the Milestone 2 backend with thoughtful UI/UX design, the platform now enables operational decision-making without requiring technical expertise or direct database access. The modular frontend architecture supports role-based access control and provides a foundation for future enhancements including mobile access, advanced filtering, and deeper analytical capabilities.

The separation of concerns across authentication, page-specific logic, visualization utilities, and API communication ensures the codebase remains maintainable and extensible as the platform evolves. Milestone 3 establishes the user-facing layer necessary for real-world operational adoption and sets the stage for Milestone 4 focus on advanced analytics, automation, and scale.

## REFERENCES

- Streamlit Documentation. <https://docs.streamlit.io/>
- Plotly Python Documentation. <https://plotly.com/python/>
- Pandas Documentation. <https://pandas.pydata.org/docs/>
- Python Requests Library. <https://requests.readthedocs.io/>
- FastAPI Documentation. <https://fastapi.tiangolo.com/>
- CSS3 Styling Guide. <https://developer.mozilla.org/en-US/docs/Web/CSS/>
- Project source code: Predictive Maintenance and Process Intelligence, Team D, Milestone 3 implementation.